An ℓ^1 Obstruction Framework for Coboundary Defects

Author: Jeremy H. Carroll **Date:** May 2026 **Version:** v5.0 Pre-Print

Abstract

Papers 000-003 established, through independent proofs at four structural levels, that within the class of edge-additive, faithful, and functorial diagnostics on coboundary defects, the diagnostic has an ℓ^1 form, generally with orbit weights unless edge-uniformity is assumed. Paper 002 proved this at the combinatorial level (coordinate additivity on finite graph cochains). Paper 001 extended it to Banach presheaves (functoriality under bounded linear maps). Paper 000 showed that projection dynamics generate defects measured by the selected observer norm, persisting at non-vanishing density under stated hypotheses. Paper 003 demonstrated that a fixed ℓ^1 cochain geometry induces the standard quotient norm on first cohomology. Alternative classes of functionals (e.g., relaxing edge-additivity to subadditivity, or faithfulness to semi-faithfulness) may yield different geometries; the uniqueness results hold precisely within the axiom class specified.

This paper synthesizes these results into a unified obstruction framework and extends them in three directions. First, we prove a contractivity statement for ℓ^1 coboundary norms under coarse-graining functors equipped with contractive pushforward assumptions (Theorem 2.1). Defects can be hidden by collapse or cancellation; no absolute no-hiding theorem is claimed. Second, we extend the framework from scalar and vector-valued presheaves to operator-valued presheaves once the local trace/Schatten-1 gauge is selected (Theorem 3.1). CPTP monotonicity by itself does not select this gauge. Third, we define the **Local ℓ^1 Primal–Dual Iterator** as a local primal–dual realization of ℓ^1 defect minimization consistent with dual feasibility constraints. Under additional operator-valued assumptions—finite trace-class stalks, trace-norm preservation, complete positivity, and compositional/naturality constraints—we show that admissible local automorphisms reduce to unitary conjugation. This supplies a conditional $U(n)/PU(n)$-type frame symmetry, not a derivation of physical gauge fields or Yang–Mills dynamics (Theorem 4.10).

We further outline, at the level of a research program, how these three extensions connect the foundational quad to subsequent applications in autopoietic structural posets (Paper 005), gauge group emergence, the fermion sign problem, and the behavior of coupling constants. All conjectural claims are explicitly marked.

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1. Introduction

1.1 The Foundational Quad

The preceding four papers establish a single structural conclusion at four levels of generality:

1. **Combinatorial (Paper 002)** [6]: On finite graph cochains, edge locality, coordinate additivity over disjoint supports, coordinate symmetry, and graph invariance select an ℓ^1 diagnostic geometry; orbit weights remain unless edge-uniformity is assumed.

2. **Functional-analytic (Paper 001)** [5]: On Banach presheaf coboundaries, edge-wise additivity, faithfulness, functoriality under bounded linear endomorphisms, and isomorphism invariance select a weighted ℓ^1 coboundary norm, with scalar uniqueness only after edge-uniformity.

3. **Dynamical (Paper 000)** [4]: Projection of linear many-body evolution onto a nonlinear manifold produces coboundary defects that (a) obstruct affine-coordinate linearity of the projected flow under the stated chart assumptions, (b) are measured by the selected ℓ^1 observer diagnostic, and (c) persist at non-vanishing density over a finite time window independent of system size.

4. **Cohomological (Paper 003)** [7]: The fixed ℓ^1 geometry on $C^1(G)$ induces the standard quotient norm Φ on $H^1(G)$. This is the quotient norm compatible with the chosen cochain geometry, not the unique possible norm on cohomology. The ℓ^2 Hodge decomposition can inflate the sum of the ℓ^1-costs of its bookkeeping components by a factor approaching 3 for the single-edge defect construction on cycle graphs.

The present paper is not a new proof of ℓ^1 rigidity. It is a synthesis: we organize the quad into a single categorical framework, extend it to new structural settings, and identify the connections to subsequent work. We emphasize that the uniqueness results hold within the precisely defined class of admissible diagnostics (edge-additive, faithful, functorial, invariant); alternative axiom classes may well select different geometries.

1.2 What This Paper Contributes

The paper contributes the following conditional results:

- **Theorem 2.1 (Functorial Aggregation Monotonicity):** Under explicit contractive pushforward assumptions, the ℓ^1 coboundary norm is monotonically non-increasing under coarse-graining functors. Defects can be hidden by collapse or cancellation; the theorem only controls when and how this decrease occurs.

- **Theorem 3.1 (Operator-Valued Extension):** When presheaf stalks are operator algebras $\mathcal{B}(\mathcal{H})$ and the local trace/Schatten-1 gauge is selected

by the observer assumptions, the edge-additive coboundary norm becomes $\ell^1(E; \|\cdot\|_1)$.

- **Theorem 4.2 (Fixed Point Characterization):** The Local ℓ^1 Primal–Dual Iterator's fixed points are precisely the ℓ^1-minimal representatives in their cohomology classes, certified by LP strong duality via divergence-free dual witnesses.
- **Theorem 4.10 (Unitary Gauge Structure):** Under the operator-valued Schatten-1 topology, preserving compositional invariance against the Iterator uniquely restricts admissible local automorphisms to unitary conjugation, corresponding to a $U(n)$ gauge structure.

1.3 Relation to the Full Program

The program proceeds in stages:

- Papers 000–003: Establish the ℓ^1 obstruction framework across combinatorial, functional, dynamical, and cohomological settings.
- Paper 004 (this work): Synthesizes and extends the framework (aggregation, operator-valued extension, dynamical structure).
- Subsequent papers: Explore applications of the framework to increasingly structured physical and computational settings, including gauge structure, fermionic systems, and coupling behavior.

1.4 Computational Companion

This paper is accompanied by illustrative Python scripts (`universal_obstruction.py` and `isotropy_emergence_primal_dual.py`) that model aspects of the framework. The former demonstrates the Distinguishability Bound Theorem (Theorem 5.1) on operator-valued presheaves; the latter illustrates the primal–dual iteration of Definition 4.1 on scalar grid graphs with visualization of defect concentration. These scripts serve as structural illustrations of the mathematical framework rather than formal physical simulations. **Note:** The primal–dual script implements a two-step Chambolle–Pock variant (omitting the divergence correction step of Definition 4.1); see Section 4.2 for the full three-step scheme.

1.5 Conventions

Norms are denoted $\|\cdot\|$ with appropriate subscripts. The Schatten-p norm on $\mathcal{B}(\mathcal{H})$ is $\|A\|_p = (\mathrm{Tr}|A|^p)^{1/p}$, where $|A| = (A^*A)^{1/2}$. The Schatten-1 norm is the trace norm $\|A\|_1 = \mathrm{Tr}|A|$. Scalar absolute values are denoted $|\cdot|$.

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2. Functorial Defect Aggregation

2.1 Setup

Let $(P, \leq)$ be a finite poset with covering relations E_{cov}, and let F be a Banach presheaf over P. A **coarse-graining functor** is a surjective order-preserving map $\pi : P \to Q$ to a smaller poset Q, together with a natural transformation that aggregates the stalks: for each $q \in Q$, the aggregated stalk is $G(q) = \bigoplus_{x \in \pi^{-1}(q)} F(x)$ with restriction maps induced from F.

The **pushforward** of a section s of F is the section $\pi_* s$ of G defined by $(\pi_* s)(q) = (s(x))_{x \in \pi^{-1}(q)}$.

2.2 Monotonicity

> **Theorem 2.1 (Functorial Aggregation Monotonicity). (Layer B)** Let $\pi : (P, F) \to (Q, G)$ be a coarse-graining as above. Assume all restriction maps are contractive with respect to the underlying Banach norms and that each coarse edge defect is obtained from the corresponding fine edge defects by a contractive aggregation map. Then
>
> $$I_Q(\pi_* s) \leq I_P(s)$$
>
> for every section s of F. Equality conditions are sufficient under the stated edge-partition model. If multiple fine edges aggregate into a single coarse edge, equality also requires no cancellation loss inside the aggregation map.

Proof. The ℓ^1 coboundary norm on P sums over all covering edges of P:

$$I_P(s) = \sum_{e \in E_{\text{cov}}(P)} \|\delta_e(s)\|$$

Under π, the edges of P partition into two classes: (a) edges $e = (a, b)$ with $\pi(a) \neq \pi(b)$, which map to covering edges of Q, and (b) edges $e = (a, b)$ with $\pi(a) = \pi(b)$, which are collapsed.

For edges of type (a), the coboundary defect on Q satisfies $\|\delta_{\pi(e)}(\pi_* s)\| \leq \|\delta_e(s)\|$ by contractivity of the induced restriction maps on the aggregated stalks.

Edges of type (b) contribute to $I_P(s)$ but not to $I_Q(\pi_* s)$, since they are no longer covering relations in Q.

Therefore:

$$I_Q(\pi_* s) \leq \sum_{e : \pi(a) \neq \pi(b)} \|\delta_e(s)\| \leq I_P(s)$$

Equality requires both that the type-(a) contractions are isometric and that all type-(b) edges have zero defect, plus no cancellation loss when aggregating fine edges to a single coarse edge. $\square$

Interpretation. Coarse-graining cannot increase the measured ℓ^1 defect under the stated contractive aggregation assumptions. It can hide defects by collapsing edges or by aggregating fine defects into a smaller coarse representative. Thus the theorem is a stability/contractivity statement, not a no-hiding theorem.

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3. Operator-Valued Presheaves and the Schatten-1 Norm

3.1 From Scalars to Operators

Papers 001 and 002 classify the ℓ^1 norm on presheaves with scalar or vector-valued stalks. In quantum many-body systems, the natural stalks are spaces of density operators on local Hilbert spaces. This section extends the classification to this setting.

Let $(P, \leq)$ be a finite poset. An **operator-valued presheaf** F over P assigns: - To each $x \in P$, a finite-dimensional Hilbert space $\mathcal{H}_x$ and the Banach space $F(x) = \mathcal{B}_1(\mathcal{H}_x)$ of trace-class operators, equipped with the Schatten-1 (trace) norm $\|A\|_1 = \mathrm{Tr}|A|$. - To each covering edge $e = (a, b)$, a completely positive trace-non-increasing restriction map $r_{b,a} : F(b) \to F(a)$ (the partial trace or a conditional expectation).

The coboundary defect at edge $e = (a, b)$ for a section $\rho = (\rho_x)_{x \in P}$ (where $\rho_x \in F(x)$ is a local density operator) is:

$$\delta_e(\rho) = r_{b,a}(\rho_b) - \rho_a \in F(a)$$

This models the failure of the local state ρ_a to match the transported or restricted state from ρ_b. In quantum-state realizations, this is a form of marginal incompatibility.

3.2 The M-Compatible Transformation Monoid

When mapping functoriality onto structured spaces (Banach presheaves), requiring invariance under *all* bounded linear maps uniquely drives the unweighted L^1 norm. However, physical state operators restrict the mapping monoid.

Let the allowable measurement transformations form an admissible monoid $M_e \subset \mathrm{End}(F(a_e))$. When M is restricted to CPTP maps, one obtains a physically motivated monotonicity class, but this restriction alone does not uniquely select the trace norm. Trace/Schatten-1 geometry must be supplied by additional operational distinguishability assumptions.

3.3 Trace-Norm Classification Extension

> **Theorem 3.1 (Operator-Valued Schatten-1 Classification).** **(Layer B)** Let ν be a seminorm on the operator-valued 1-cochain space $C^1(P, F) = \bigoplus_{e \in E_{\mathrm{cov}}} \mathcal{B}_1(\mathcal{H}_{a_e})$ satisfying:

(P1) Edge-wise additivity: $\nu((\delta_e)_e) = \sum_e \nu_e(\delta_e)$

(P2) Faithfulness: $\nu_e(\delta) = 0 \iff \delta = 0$

(P3) Functoriality: $\nu_e(\Phi(\delta)) \leq \|\Phi\|_{\mathrm{op}} \cdot \nu_e(\delta)$ for every bounded linear map $\Phi : \mathcal{B}_1(\mathcal{H}) \to \mathcal{B}_1(\mathcal{H})$

(P4) Isomorphism invariance

Then $\nu_e(\delta) = w_e \cdot \|\delta\|_1$ for some $w_e > 0$, where $\|\cdot\|_1$ is the Schatten-1 (trace) norm. If (P4) holds, $w_e = \alpha$ for all e.

Proof sketch. The argument follows the structure of Paper 001, Theorem 2.2. The key step is the direction-independence lemma: for any two operators $A, B \in \mathcal{B}_1(\mathcal{H})$ with $\|A\|_1 = \|B\|_1 \neq 0$, we must show $\nu_e(A) = \nu_e(B)$.

By the operator-valued Hahn–Banach theorem (a consequence of the duality $\mathcal{B}_1(\mathcal{H})^* = \mathcal{B}(\mathcal{H})$), for any A with $\|A\|_1 = 1$, there exists $X \in \mathcal{B}(\mathcal{H})$ with $\|X\|_{\mathrm{op}} = 1$ and $\mathrm{Tr}(X^*A) = 1$. Define the rank-one map $\Phi : \mathcal{B}_1(\mathcal{H}) \to \mathcal{B}_1(\mathcal{H})$ by $\Phi(C) = \mathrm{Tr}(X^*C) \cdot B/\|B\|_1$, which is bounded with operator norm ≤ 1; the extension to complete boundedness follows from standard results on trace-class duality. Then $\Phi(A) = B$, and (P3) gives $\nu_e(B) \leq \nu_e(A)$. By symmetry, equality follows.

The remainder proceeds identically to the scalar case: direction independence plus homogeneity forces $\nu_e = w_e\|\cdot\|_1$, and (P4) forces uniform weights. □

Remark 3.1 (Physical interpretation). Theorem 3.1 is a maximal bounded-linear classification on the trace-class Banach space. It should not be read as saying CPTP monotonicity alone forces the trace norm. In quantum applications, trace distance is selected only after adding operational distinguishability assumptions such as classical total-variation reduction and binary discrimination duality.

3.3 Structural Consequences

Quantum marginal incompatibility. When ρ is a global quantum state and $\Pi(\rho) = \bigotimes_x \mathrm{Tr}_{\bar{x}}(\rho)$ is the product-state retraction, the coboundary defect $\delta_e(\Pi(\rho))$ measures the entanglement across edge e in trace norm. Theorem 3.1 shows uniqueness only under the maximal bounded-linear functoriality assumption stated in (P3). Under the physically narrower CPTP monoid, trace norm must be separately selected by operational distinguishability assumptions; CPTP monotonicity alone is not a uniqueness theorem.

Entanglement constraints. The edge-wise additivity (P1) combined with the trace norm structure places strict bounds on the total defect across all edges sharing a vertex. This additive limit is formally compatible with monogamy-type intuitions, but no monogamy theorem is proved here.

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4. The Local ℓ^1 Primal–Dual Iterator

4.1 Discarding the ℓ^2 Laplacian

Classical formulations utilize the ℓ^2 Sheaf Laplacian ($\Delta_{\mathcal{F}} = d^*d$) as an analytically convenient smoothing operator to define local dynamics. Paper 003 shows that ℓ^2 Hodge representatives can inflate ℓ^1 cost in a specific cycle-graph construction. If the observer has selected the ℓ^1 quotient norm as the acceptance geometry, an ℓ^2 Laplacian flow is therefore an auxiliary algorithmic choice, not the native variational geometry.

4.2 The Local ℓ^1 Primal–Dual Iterator (Discrete Form)

The differential inclusion associated with minimizing $\|\omega - d_0 s\|_1$ provides a formal characterization of ℓ^1 residual repair, but does not by itself define a constructive or local dynamical system.

We therefore introduce an explicit **discrete primal–dual iteration** whose fixed points coincide with ℓ^1-minimal representatives. Among local update rules consistent with the ℓ^1 variational principle and dual feasibility constraints ($d_0^*\phi = 0$, $\|\phi\|_\infty \leq 1$), this primal–dual scheme provides a natural realization: the dual step enforces sign alignment, the divergence correction enforces conservation, and the primal step follows the resulting gradient. The fixed-point set is determined by the LP optimality conditions (Theorem 4.2) and is shared by any correct solver for the ℓ^1 minimization problem; the specific three-step decomposition is chosen for its physical interpretability and locality, not claimed to be unique among all possible algorithms.

Definition 4.1 (Local ℓ^1 Primal–Dual Iterator). Let (P, F) be a finite presheaf over a poset with covering edges E_{cov}. Fix an observed edge field $\omega \in C^1(P, F)$ and let $s^k \in C^0(P, F)$ be a section at iteration k. Define the repair residual:

$$\delta^k = \omega - d_0 s^k \in C^1(P, F).$$

The special case $\omega = 0$ is a section-consistency energy, not a nontrivial cohomology quotient.

Introduce dual variables $\phi^k \in C^1(P, F)$, initialized arbitrarily with $\|\phi^0\|_\infty \leq 1$.

For step sizes $\sigma, \tau, \eta > 0$, define:

(1) Dual ascent (edge-local)

$$\phi^{k+\frac{1}{2}}(e) = \text{clip}_{[-1,1]}\left(\phi^k(e) + \sigma \cdot \delta^k(e)\right)$$

(2) Divergence correction (vertex-local)

$$\phi^{k+1} = \phi^{k+\frac{1}{2}} - \tau \cdot d_0(d_0^*\phi^{k+\frac{1}{2}})$$

(3) Primal update (vertex-local)

$$s^{k+1} = s^k + \eta \cdot d_0^* \phi^{k+1}$$

Interpretation: * Step (1) aligns dual variables with the local defect (sign selection). * Step (2) enforces the approximate divergence-free constraint $d_0^*\phi \approx 0$. * Step (3) updates the section to reduce the repair residual.

All operations are strictly local: edge updates depend only on edge endpoints, and vertex updates depend only on adjacent edges.

Proposition 4.1 (Self-Consistent Defect Resolution). The Local ℓ^1 Primal–Dual Iterator defines a closed local system in which: 1. Defects generate dual responses (Step 1). 2. Dual responses enforce global consistency (Step 2). 3. Consistency feeds back to reduce defects (Step 3).

The system requires no global solver and maintains only local interactions. Under appropriate step-size conditions (which depend on the spectral radius of $d_0 d_0^*$ and the interaction between the three steps; see Remark below), convergence is conjectured; the scheme is consistent with known primal–dual methods but differs due to the intermediate correction step. Full convergence analysis, including explicit step-size bounds, is deferred to subsequent work.

Remark (Step-size conditions). The three-step scheme of Definition 4.1 is structurally related to Chambolle–Pock primal–dual splitting [8], but the intermediate divergence correction (Step 2) makes it a non-standard variant. The standard Chambolle–Pock step-size condition $\sigma\tau\|d_0\|_{\mathrm{op}}^2 < 1$ does not directly apply without accounting for the correction step. Establishing the precise step-size regime for this three-step scheme is an open problem.

Theorem 4.2 (KKT Fixed Point Characterization). (Layer B) Let (s^*, ϕ^*) be a fixed point of the Local ℓ^1 Primal–Dual Iterator that also satisfies the KKT feasibility conditions for the quotient problem. Then:

$$\omega - d_0 s^* = \delta_{\mathrm{irr}}$$

where δ_{irr} is an ℓ^1-minimal representative of the quotient class $[\omega]$. Moreover, there exists a dual witness ϕ^* such that: * $\phi^* \in \partial\|\delta_{\mathrm{irr}}\|_1$ * $d_0^*\phi^* = 0$ * $\langle \phi^*, \delta_{\mathrm{irr}} \rangle = \|\delta_{\mathrm{irr}}\|_1$

At the fixed point, the system stabilizes at the ℓ^1 primal-dual decomposition $\omega = d_0 s^* + \delta_{\mathrm{irr}}$. The ℓ^1-minimality follows from LP strong duality (Paper 003, Theorem 9a.6; cf. [8]): the dual feasibility condition $d_0^*\phi^* = 0$ together with the complementary slackness condition $\langle \phi^*, \delta_{\mathrm{irr}} \rangle = \|\delta_{\mathrm{irr}}\|_1$ certify primal optimality. We emphasize that Theorem 4.2 characterizes the structure of fixed points; convergence of the iteration to a fixed point is conjectured (Conjecture 4.3) but not proved here.

Remark. The corresponding continuous-time inclusion for $\|\omega - d_0 s\|_1$ is formally recovered as the continuous-time limit of this discrete iterator.

Conjecture 4.3 (Local ℓ^1 Defect Descent). (Layer C) Let s^k evolve under the Local ℓ^1 Primal–Dual Iterator with sufficiently small step sizes $\sigma, \tau, \eta > 0$. Then the ℓ^1 residual functional $I(s^k) := \|\omega - d_0 s^k\|_1$ satisfies:

$$I(s^{k+1}) \leq I(s^k)$$

with strict inequality unless s^k is a fixed point.

Proof Sketch. 1. **Dual alignment increases pairing:** From Step (1), $\phi^{k+\frac{1}{2}} \approx \operatorname{sign}(\delta^k)$, implying $\langle \phi^{k+\frac{1}{2}}, \delta^k \rangle \approx \|\delta^k\|_1$, with clipping ensuring $\|\phi^{k+\frac{1}{2}}\|_\infty \leq 1$. 2. **Divergence correction preserves feasibility:** Step (2) reduces $d_0^* \phi$ while keeping ϕ approximately within the feasible dual set. 3. **Primal update reduces defect:** Step (3) evolves the residual as $\delta^{k+1} = \omega - d_0 s^{k+1} = \delta^k - \eta d_0 d_0^* \phi^{k+1}$. 4. **Energy decrease (incomplete):** One expects $\|\delta^{k+1}\|_1 \leq \|\delta^k\|_1 - \eta \langle \phi^{k+1}, d_0 d_0^* \phi^{k+1} \rangle + O(\eta^2)$. Since $d_0 d_0^*$ is positive semidefinite and ϕ^{k+1} aligns with δ^k, descent would follow. However, the ℓ^1 norm is non-smooth, and the standard convexity inequality provides a lower bound, not the upper bound needed here. A rigorous descent proof requires either a Lyapunov function argument (e.g., showing the primal-dual gap decreases monotonically) or appeal to the theory of monotone operator splitting with explicit step-size conditions for this three-step variant. This gap is the reason we classify descent as Layer C rather than Layer B. □

Interpretation. The Local ℓ^1 Primal–Dual Iterator is designed as a **descent process on the ℓ^1 defect functional**. Each iteration aligns dual variables with local defect structure, enforces global consistency via divergence correction, and updates the primal state to reduce the total defect. The system is designed to act as a locally driven global optimizer, with fixed points characterized by ℓ^1 minimality; convergence remains an open problem.

Conjecture 4.4 (Local Edge-Wise Descent Heuristic). (Layer C) Let s^k evolve under the Local ℓ^1 Primal–Dual Iterator. For each edge $e = (a, b)$, define the local defect $\delta^k(e) = r_{b,a}(s^k(b)) - s^k(a)$. For sufficiently small step sizes one expects:

$$|\delta^{k+1}(e)| \leq |\delta^k(e)| + \sum_{e' \sim e} \epsilon_{e,e'}$$

where $e' \sim e$ are edges sharing a vertex with e, and $\epsilon_{e,e'}$ are higher-order coupling terms bounded at $O(\eta^2)$.

Interpretation. Each edge attempts to reduce its own defect using only local information, experiencing bounded interference from neighboring edges. This motivates the system as a distributed local descent scheme, but a rigorous monotonic descent theorem requires a Lyapunov or operator-splitting proof and is not established here.

Heuristic. The update at a vertex depends only on adjacent edges. For an edge $e = (a, b)$, the defect evolves as $\delta^{k+1}(e) = \delta^k(e) - \eta\big(\sum_{e' \ni b} \phi(e') - \sum_{e' \ni a} \phi(e')\big)$. This decomposes into a self-contribution from e and cross-contributions from

neighboring edges. The clipping bound controls the size of those terms, but it does not by itself prove monotonic local decrease.

Heuristic 4.2 (Cycle Concentration Mechanism) (Layer C). When the quotient class is nonzero and the iterator behaves like a correct local solver for the ℓ^1 quotient problem, residual defects are expected to concentrate on representatives compatible with the cycle-space constraints. This is a mechanism hypothesis, not a convergence theorem.

(This outlines the anticipated mechanism for structure formation: local incompatible constraints cannot all be satisfied simultaneously at vertices, leading to expected concentration on topological cycles).

4.3 Autopoietic Structures and Cycle Stability

We now formalize the conditions under which topological defects persist as stable macroscopic features rather than dissolving under the optimization flow.

Definition 4.2 (Cycle-Supported Defect). A defect field $\delta \in C^1$ is *cycle-supported* if its support lies entirely on a set of edges forming a topological cycle $C \subseteq E$, and satisfies $d_0^* \phi = 0$ for some $\phi \in \partial \|\delta\|_1$. Equivalently, the defect has no net divergence at any vertex and is natively in the dual-feasible space.

Proposition 4.5 (Certified Cycle-Supported Fixed Points). Assume (δ^k, ϕ^k) satisfies the fixed-point equations of Definition 4.1, with $\phi^k \in \partial \|\delta^k\|_1$, $d_0^* \phi^k = 0$, and the clipping/divergence-correction steps leave ϕ^k invariant. Then the primal residual remains fixed and δ^k is ℓ^1-minimal in its quotient class.

Proof Sketch. At the discrete update, $d_0^* \phi^k = 0$ implies the primal update vanishes ($s^{k+1} = s^k \Rightarrow \delta^{k+1} = \delta^k$). Because ϕ^k strictly lies in the subdifferential and is divergence-free, the configuration is ℓ^1-minimal. Assuming the clipping and correction steps leave the dual variable invariant, neither primal nor dual state adjusts. □

Interpretation. Cycle-supported defects with divergence-free invariant dual certificates are fixed points under local optimization: no local iteration can reduce them because cancellation strictly requires divergence, which cycles lack. Consequently, system dynamics trap unresolvable frustration specifically on cycles, acting as the system's topological attractors.

Definition 4.3 (Autopoietic Structure). A defect configuration is a formal *Autopoietic Structure* (a proto-object) if it is: 1. ℓ^1-minimal in its cohomology class 2. Supported structurally on a geometric cycle 3. Strictly invariant under the continued Local ℓ^1 Primal–Dual Iterator flow.

Corollary 4.1 (Tree Exactness). On a tree, every edge defect is exact at the quotient level: $C^1 / \operatorname{im}(d_0) = 0$. Therefore no cohomological survivor exists. Whether a particular local iterator removes the representative defect depends on its convergence behavior.

4.4 Cycle Interaction and the ℓ^1 Potential

Once certified cycle-supported configurations are present, their combined quotient class may interact through shared constraints across macroscopic vertices.

Heuristic 4.6 (Cycle Interaction Principle). (Layer C) Let two stable cycle-supported defects $\delta^{(1)}, \delta^{(2)}$ evolve together under the Local ℓ^1 Primal–Dual Iterator, evaluated collectively as $\delta = \delta^{(1)} + \delta^{(2)}$. 1. **Disjoint Case:** If $\mathrm{supp}(\delta^{(1)}) \cap \mathrm{supp}(\delta^{(2)}) = \emptyset$, then δ remains a stable fixed point. Disjoint cycles assert no interactive coupling constraint. 2. **Overlapping Case:** If the cycles share vertices, the combined configuration δ is generically (i.e., outside a measure-zero set in the space of defect assignments) not ℓ^1-minimal. The combined dual may fail feasibility either through divergence ($d_0^*(\phi^{(1)} + \phi^{(2)}) \neq 0$ at shared vertices) or through norm-bound violation ($\|\phi^{(1)} + \phi^{(2)}\|_\infty > 1$ at shared edges). In either case, the fixed-point condition may fail, so a descent-capable iterator would be expected to re-optimize the combined configuration $\delta \to \delta_{\mathrm{new}}$ with strictly lower total ℓ^1 cost.

Mechanism. Where interaction vertices are shared, the independent cycles induce competing dual flows $\phi^{(1)}, \phi^{(2)}$. The local iterator evaluates the simple scalar sum $d_0^*(\phi^{(1)} + \phi^{(2)}) \neq 0$. The resulting newly non-zero divergence breaks the fixed-point condition, reactivating the iterative physical flow to minimize the combined macroscopic defect. Continuous interaction evaluates not as an extrinsic phenomenological "force," but strictly as structural re-optimization under logically shared constraints.

Definition 4.4 (ℓ^1 Interaction Functional). We define the topological interaction potential between adjacent stable cycles on cohomology classes:

$$\mathcal{I}([\delta^{(1)}], [\delta^{(2)}]) = \Phi([\delta^{(1)} + \delta^{(2)}]) - \big(\Phi([\delta^{(1)}]) + \Phi([\delta^{(2)}])\big)$$

where $\Phi([\delta]) = \min_f \|\delta - d_0 f\|_1$ is the ℓ^1 quotient norm from Paper 003. Since Φ is a norm on quotient classes, the triangle inequality gives

$$\mathcal{I}([\delta^{(1)}], [\delta^{(2)}]) \leq 0.$$

Therefore this functional measures only subadditive synergy/cancellation. It cannot diagnose a positive repulsive or superadditive interaction. Any such effect must be represented by a different functional, for example one depending on constrained representatives, structural lift costs, or forbidden shared gauge degrees of freedom.

This interaction functional enables the following structural properties: * **Subadditive regime ($\mathcal{I} < 0$):** The joint gauge optimization achieves greater cancellation than separate optimizations, pulling the configuration tighter together. * **Fusion:** Two overlapping cycles merge into a single extended defect cycle if $\Phi([\delta_{\mathrm{merged}}]) < \Phi([\delta^{(1)}]) + \Phi([\delta^{(2)}])$. * **Annihilation:** If $[\delta^{(1)}] = -[\delta^{(2)}]$, the interaction enables complete phase cancellation $[\delta] \to 0$.

4.5 Defect Propagation and Causal Signals

To complete the physical picture of localized dynamics, we must specify how disturbances propagate through the ℓ^1 lattice.

Definition 4.5 (Local Disturbance). Let s be a stable configuration with defect $\delta = \delta_{\text{irr}}$. A *local disturbance* at vertex x is a perturbation $s(x) \mapsto s(x) + \epsilon$ inducing a defect change $\delta \mapsto \delta + \Delta\delta$ supported on edges incident to x.

Proposition 4.7 (One-Step Residual Update Identity). (Layer B) Under the Local ℓ^1 Primal–Dual Iterator, a local disturbance propagates along adjacent edges in a single step according to the algebraic identity:

$$\delta^{k+1} = \delta^k - \eta\, d_0 d_0^* \phi^k$$

with dual certificate $\phi^k \in \partial\|\delta^k\|_1$. This is a one-step algebraic identity, not a proof of stable signal behavior or physical propagation.

Mechanism. At the disturbed vertex, consistency breaks, producing localized divergence ($d_0^*\phi \neq 0$). The dual update aligns with the defect sign structure; divergence correction redistributes the imbalance subject to $\|\phi\|_\infty \leq 1$.

Interpretation. The propagation operator $d_0 d_0^*$ acts as a discrete Laplacian, but driven by the ℓ^1 subgradient rather than ℓ^2 diffusion. This distinction is fundamental: ℓ^2 diffusion spreads mass uniformly and destroys structure, while ℓ^1 propagation is *edge-selective*, preferring minimal paths and concentrating signal flow along sharp boundaries.

Corollary 4.2 (Finite-Speed Propagation). A disturbance at vertex x affects only vertices within graph distance k after k iterations.

This enforces graph-local influence structure in the weak sense: local updates dictate finite propagation speed, eliminating instantaneous non-local effects. The iteration index k serves as the intrinsic time parameter, corresponding to descent in the ℓ^1 action rather than an externally imposed clock. However, this graph-local influence does not by itself imply an invariant propagation speed, Lorentz symmetry, or observer-independent frame structure; propagation speed depends on the local graph connectivity and step-size parameters. The framework derives interaction ordering from variational descent; metric spacetime structure remains an open emergence problem.

(Interaction with Cycles). When a propagating disturbance encounters a certified cycle-supported fixed point (Proposition 4.5), the cycle resists absorption due to its divergence-free certification. The disturbance routes *around* the cycle or modifies only the adjacent lattice. These cycles therefore act as topological obstacles to propagation—providing interaction between propagating disturbances and certified residual cycles derived purely from ℓ^1 constraint logic.

Remark 4.3 (Cone Structure from Dual Feasibility). The dual feasibility constraint $\|\phi\|_\infty \leq 1$ imposes a bounded flux per edge per iteration. Combined with locality, this induces a cone of admissible influence: after k steps from

vertex x, only vertices within the set $\mathcal{C}(x,k) = \{y \mid d_G(x,y) \leq k\}$ can be affected, and the magnitude of influence is bounded by the cumulative flux capacity along any path. Because ℓ^1 dynamics prefer minimal-support flows (sparse paths), propagation concentrates along lowest-cost routes rather than spreading isotropically. The defect field itself naturally defines a path cost $d_\delta(x,y) = \inf_{\gamma:x\to y} \sum_{e\in\gamma} |\delta(e)|$, yielding an emergent metric structure in which geometry depends on the field configuration. This provides a structural basis for causal geometry arising from the dynamics, rather than being imposed a priori. However, this cone and metric are graph-dependent and generically anisotropic; whether they can yield isotropic large-scale geometry under aggregation is an open problem (see Section 6.4).

4.6 The Continuum Limit: Total Variation Flow

The discrete ℓ^1 rules suggest a formal continuum scaling. This section records the expected TV-flow limit under standard assumptions from graph-to-continuum variational convergence. No new Γ-convergence or minimizing-movement theorem is proved here. The logical chain is: ℓ^1 can favor sparse or measure-concentrated representatives under appropriate compactness and scaling hypotheses; measure-valued gradients live in the space of functions of bounded variation (BV); and the natural variational functional on BV is total variation. The ℓ^1 coboundary structure therefore is consistent with total variation–type continuum functionals under standard scaling limits. The precise isotropy of the limiting functional depends on the symmetry of the underlying discretization: anisotropic or crystalline graph sequences yield anisotropic TV, while isotropic refinement yields the standard (Euclidean) total variation. The continuum geometric structure is inherited from the embedding of the graph sequence; the ℓ^1 framework determines the variational form, while the ambient geometry determines its metric structure.

Schema 4.8 (Continuum TV Limit Analogy). (Layer C — Formal Limit) Let G_h be a sequence of graphs with uniformly bounded degree admitting a consistent geometric embedding in a domain $\Omega \subset \mathbb{R}^d$, with appropriately scaled edge weights and mesh spacing $h \to 0$. Under isotropic refinement, the rescaling $I_h(s_h) = \sum_e |\delta_h(e)| \to \int_\Omega |\nabla s(x)|\, dx$ admits total variation flow as the canonical local continuum limit, and the Local ℓ^1 Primal–Dual Iterator is formally consistent with a minimizing-movement discretization of this flow:

$$\partial_t s(x,t) \in -\partial \int_\Omega |\nabla s(x,t)|\, dx$$

Where the field is smooth, this reduces to:

$$\partial_t s = \nabla \cdot \left(\frac{\nabla s}{|\nabla s|} \right)$$

Interpretation. The iterator defines an intrinsic time parameter corresponding to descent in the ℓ^1 action, yielding a minimizing-movement interpretation of

evolution (in the sense of De Giorgi), rather than a direct time discretization of the PDE. In formal continuum scaling, the limiting variational structure is consistent with total-variation flow, a nonlinear edge-preserving evolution in the scalar case. This establishes a fundamental structural contrast:

1. **Classical ℓ^2 Diffusion:** $\partial_t s = \Delta s$. Isotropically smooths all gradients, destroying object structure toward featureless equilibrium.
2. **Autopoietic ℓ^1 Flow:** $\partial_t s = \nabla \cdot (\nabla s / |\nabla s|)$. Preserves step-edges, concentrates gradients into sharp topological boundaries, and drives geometry under mean curvature.

Crucially, the dynamics do not eliminate structure globally; they concentrate irreducible defect onto stable sets (cycles in the discrete setting, interfaces in the continuum), preserving topological information while reducing total variation. Discrete cycles characterizing autopoietic stable objects converge to sharp level-set boundaries (codimension-1 manifolds) enclosing persistent domains, consistent with known graph-to-TV Γ-convergence results in the variational literature (cf. Chambolle 2004, van Gennip–Bertozzi 2012). In the operator-valued case, the Schatten-1 functional $\int_\Omega \|\nabla \rho(x)\|_1 \, dx$ suggests a formally analogous structure, though a rigorous continuum theory for operator-valued total variation remains open and constitutes a nontrivial extension beyond the scalar setting.

4.7 Fiber-Wise Symmetry and $U(n)$ Gauge Fields

Because the Local ℓ^1 Primal–Dual Iterator continuously strips away exact artifacts to isolate topological frustration, it enforces strict constraints on the emergent symmetry groups. We impose a single mathematically necessary constraint ensuring that optimization logic remains stable across equivalent observers:

Axiom (Local Compositional Invariance). The absolute ℓ^1 defect magnitude must remain invariant under local relabelings of internal states. Let $s(x) \mapsto g_x s(x)$. Preserving the coboundary norm requires:

$$\|\delta_e(g \cdot s)\|_1 = \|\delta_e(s)\|_1$$

This invariance immediately forces the naturality constraint $r_{b,a} \circ g_b = g_a \circ r_{b,a}$.

Lemma 4.9 (Trace-Norm Automorphisms). Any surjective trace-norm preserving, completely positive, compositional automorphism of $\mathcal{B}_1(\mathcal{H})$ is unitary conjugation. *Proof Sketch.* By the Kadison isometry theorem [9] and its extension to Schatten ideals by Sourour [10], every surjective linear isometry of $\mathcal{B}_1(\mathbb{C}^n)$ has the form $A \mapsto UAV$ or $A \mapsto UA^TV$ for unitaries U, V. The requirement of complete positivity excludes the transpose (which is positive but not completely positive for $n \geq 2$, as its Choi matrix has a negative eigenvalue). The requirement of compositional consistency over the presheaf restriction maps ($g_a \circ r_{b,a} = r_{b,a} \circ g_b$) forces $V = U^*$: when $r_{b,a}$ includes partial traces, the equation $U_a \operatorname{Tr}_{\bar{a}}(\rho) V_a = \operatorname{Tr}_{\bar{a}}(U_b \rho V_b)$ for all ρ requires $V_a = U_a^*$ (see, e.g., Watrous, *The Theory of Quantum*

Information, Proposition 2.24). The admissible transformations reduce to unitary conjugation $A \mapsto UAU^*$. $\square$

Theorem 4.10 (Unitary Gauge Structure). (Layer B) When stalks are finite-dimensional operator algebras $F(x) = \mathcal{B}_1(\mathbb{C}^n)$ evaluated under the Schatten-1 trace norm (Theorem 3.1), the local transformations satisfying compositional invariance restrict admissible local automorphisms to unitary conjugation, corresponding to a $U(n)$ gauge structure.

1. By Lemma 4.9, the automorphisms g_x must be trace-norm isometries acting via unitary conjugation: $g_x(A) = U_x A U_x^*$.
2. The presheaf restriction maps $r_{b,a}$ behave as **discrete parallel transport**, defining geometric paths across the tensor network.
3. The optimal terminal defect δ_{irr} acts as the unresolved quotient defect, formally analogous to curvature/holonomy constrained within the graph.

Under these additional operator-valued assumptions, the admissible local frame changes are unitary conjugations. This supplies a conditional $U(n)/PU(n)$-type gauge symmetry for the operator-valued realization. It does not derive a physical gauge field, Standard Model gauge group, or Yang–Mills dynamics.

Remark (U(n) vs. PU(n)). The gauge group acts by conjugation $A \mapsto UAU^*$, under which the center $U(1) \subset U(n)$ acts trivially (scalar phase cancels). The group that acts faithfully on the stalks is therefore $PU(n) = U(n)/U(1)$. We write $U(n)$ to denote the group of local frame changes (the assignment $x \mapsto U_x$), which is the standard convention in lattice gauge theory where the $U(1)$ phase carries physical meaning as the global gauge choice. Complete positivity is an additional physical axiom motivated by quantum mechanics (quantum channels must be CP); it is not forced by the ℓ^1 structure alone.

Establishing subgroups $SU(3) \times SU(2) \times U(1)$ requires explicit constraints (rank, phase, symmetry breaking) that descend from specific bounded dual topologies during combinatorial lattice scaling.

4.8 Geometry Emergence: Manifolds from Defects

The formal continuum analogy suggests a geometric consequence of ℓ^1 structure.

Schema 4.11 (Interface Emergence Analogy). (Layer C) In the continuum limit of the ℓ^1 defect functional $E(s) = \int_\Omega |\nabla s(x)|\, dx$, under standard BV regularity assumptions, the support of the singular part of ∇s concentrates on codimension-1 sets $\Sigma \subset \Omega$, which evolve under total variation flow by mean curvature.

Proof Sketch. Total variation minimization selects functions of bounded variation whose gradients are Radon measures. By the structure theorem for BV functions (under appropriate regularity), the singular part of ∇s concentrates on rectifiable sets of codimension 1. Under TV flow, these sets evolve by mean curvature

motion (established in geometric measure theory by the work of Evans–Spruck and Chen–Giga–Goto). □

Interpretation. The ℓ^1 framework produces a *stratified geometry* in which the field is constant almost everywhere, with all structure concentrated on evolving codimension-1 manifolds Σ. The codimension-1 nature of emergent structures follows from the measure-minimizing property of total variation, which penalizes surface measure rather than volume. The emergent interfaces are not static regularity artifacts but dynamically interacting structures governed by a variational principle and conservation laws (Theorems 4.14–4.15), elevating them from mere features of BV regularity to carriers of physical structure. In this framework, geometry is encoded variationally through the ℓ^1 action rather than through a pre-specified metric tensor. The discrete-to-continuum correspondence is direct:

Discrete	Continuum
Stable cycles	Codimension-1 interfaces Σ
Cycle stability (Thm 4.5)	Interface persistence under curvature flow
Cycle interaction (Thm 4.6)	Interface merging, splitting, annihilation
Tree instability (Cor 4.1)	Smooth regions carry no structure

4.9 Emergent Connection–Curvature–Holonomy Structure

The dual variables and cycle defects of the ℓ^1 framework naturally organize into a connection-like geometric structure.

Definition 4.6 (ℓ^1 Holonomy). For a cycle C in the graph, define the ℓ^1 holonomy:

$$\mathrm{Hol}(C) := \sum_{e \in C} \delta_{\mathrm{irr}}(e)$$

Since exact parts cancel around any cycle ($\sum_{e \in C} d_0 f = 0$), holonomy depends only on the cohomology class $[\delta]$.

Definition 4.7 (ℓ^1 Connection). Edge restriction maps $r_{b,a}$ define local transport rules. The defect $\delta_e = r_{b,a}(s(b)) - s(a)$ measures transport failure. The connection is the collection of local transport operators; the defect is its curvature.

Definition 4.8 (ℓ^1 Curvature). For a cycle C, define $F(C) := \mathrm{Hol}(C)$. Curvature is the failure of consistent transport around a closed loop—structurally identical to gauge curvature.

Theorem 4.12 (Dual Field as Conserved Flux). (Layer B) For the finite graph dual condition, the dual variable ϕ satisfies:

$$\|\phi\|_\infty \leq 1, \quad d_0^* \phi = 0$$

Schema 4.12 (Continuum Flux Analogy). (Layer C) In the formal continuum analogy, this corresponds to a divergence-free unit flux field:

$$|\phi(x)| \leq 1, \quad \nabla \cdot \phi = 0$$

aligned with defect gradients and supported on the emergent interfaces Σ.

Proof Sketch. The dual optimality condition $d_0^* \phi = 0$ becomes $\nabla \cdot \phi = 0$ in the continuum. The constraint $\|\phi\|_\infty \leq 1$ persists under scaling. Since ∇s concentrates on Σ, the alignment condition $\phi \in \partial|\nabla s|$ forces ϕ to be supported on Σ with ϕ parallel to the normal direction. □

Schema 4.13 (Connection–Curvature–Holonomy Analogy). (Layer C) In the ℓ^1 formal schema: 1. Edge defects define a *connection* via local transport inconsistency. 2. Cycle defects define *holonomy* as irreducible accumulated mismatch. 3. Holonomy defines *curvature*. 4. Dual variables define *conserved flux fields* compatible with this structure.

We emphasize that this construction does not assume a Lie group-valued connection or a principal bundle; rather, the connection–curvature–holonomy structure arises abstractly from defect transport on presheaf stalks. The parallel to standard gauge theory is structural, not definitional. Combined with the $U(n)$ result (Theorem 4.10), in the operator-valued setting the connection acts on trace-class stalks with unitary gauge symmetry, and holonomy measures the accumulated Schatten-1 mismatch around closed loops.

4.10 The ℓ^1 Gauge Action and Variational Principle

We now promote the defect functional to a gauge-invariant action.

Definition 4.9 (ℓ^1 Gauge Action). Fix an observed edge field $\omega \in C^1$. Define the **section-level residual action**:

$$\mathcal{S}(s; \omega) := \|\omega - d_0 s\|_1$$

This is not gauge-invariant as a function of a chosen section. The **gauge-invariant action** is the quotient norm of the observed field:

$$\bar{\mathcal{S}}([\omega]) := \Phi([\omega]) = \min_f \|\omega - d_0 f\|_1$$

which depends only on the quotient class $[\omega] \in C^1/\operatorname{im}(d_0)$. Physical configurations minimize $\bar{\mathcal{S}}$; the section-level action $\mathcal{S}(s; \omega)$ is an upper bound for $\bar{\mathcal{S}}([\omega])$.

Theorem 4.14 (ℓ^1 Variational Principle). (Layer B) For a fixed observed field ω, distinguished quotient representatives are minimizers of the gauge-invariant action within the quotient class:

$$\delta_{\text{irr}} \in \arg\min_f \|\omega - d_0 f\|_1.$$

For a minimizing representative $\delta = \omega - d_0 s$, the KKT condition is:

$$d_0^* \phi = 0, \quad \phi \in \partial \|\delta\|_1$$

That is: the field equation *is* the divergence-free dual condition. The divergence-free dual condition is the finite-dimensional analogue of a conservation equation.

Curvature Formulation. When the irreducible defect δ_{irr} is supported on edge-disjoint cycles $\{C_i\}$, the gauge-invariant action rewrites as:

$$\bar{\mathcal{S}} = \sum_i |\text{Hol}(C_i)|$$

For general defect configurations on overlapping cycles, this decomposition is a lower bound; the exact action requires solving the ℓ^1 quotient norm optimization. The cycle-basis representation is the structural ℓ^1 analogue of the Yang–Mills action $\sum \|F\|^2$, with ℓ^1 replacing ℓ^2.

Dual Formulation. By LP duality:

$$\bar{\mathcal{S}}([\omega]) = \max_{\phi:\, d_0^* \phi = 0,\, \|\phi\|_\infty \le 1} \langle \phi, \omega \rangle$$

The action equals the maximum flux compatible with divergence-free and boundedness constraints. This primal–dual identity—action equals flux capacity—is the structural core of the framework.

Continuum Action. Under the scaling of Theorem 4.8:

$$\mathcal{S}(s; \omega) = \int_\Omega |\omega - \nabla s| \, dx$$

whose variation recovers TV flow, confirming consistency with the iterator dynamics.

4.11 Symmetry and Conservation: ℓ^1 Noether Correspondence

Definition 4.10 (Symmetry of the ℓ^1 Action). A transformation T acting on sections and observed fields is a *symmetry* if $\|\omega' - d_0(Ts)\|_1 = \|\omega - d_0 s\|_1$ for the induced transformed field ω'. Examples include compatible additive gauge shifts, unitary conjugation (operator-valued case, Theorem 4.10), and graph automorphisms.

Proposition 4.15 (Finite-Dimensional Subgradient Conservation Condition). (Layer B) Let T_ϵ be a continuous one-parameter family of symmetries of the ℓ^1 action. Then along any solution of the ℓ^1 variational principle, there exists a dual field ϕ such that:

$$d_0^* \phi = 0 \quad \text{and} \quad \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} \|\omega_\epsilon - d_0(T_\epsilon s)\|_1 = 0$$

Proof Sketch. Action invariance gives $\frac{d}{d\epsilon}\mathcal{S}(T_\epsilon s;\omega_\epsilon) = 0$. By the subgradient structure, $\delta\mathcal{S} = \langle\phi, \delta(\omega - d_0 s)\rangle$ for $\phi \in \partial\|\delta\|_1$. For section variations with fixed ω, the stationarity condition integrates by parts (discrete transpose) to $\langle d_0^*\phi, \dot{s}\rangle = 0$. Since this holds for all admissible variations $\dot{s}$, we obtain $d_0^*\phi = 0$. **Regularity caveat:** The ℓ^1 norm is not differentiable at points where $\delta_e = 0$ for some edges. At such points, the subdifferential $\partial\|\delta\|_1$ is multi-valued (the dual variable $\phi(e) \in [-1, 1]$ is not uniquely determined). The conservation law holds for every subgradient selection $\phi \in \partial\|\delta\|_1$, but the conserved dual field is not unique at non-smooth configurations. □

Interpretation. At a minimizer of the quotient action, KKT dual feasibility gives $d_0^*\phi = 0$. This is formally analogous to a conservation law. Symmetries of the ℓ^1 action generate divergence-free dual fields—conserved flux. The correspondence is structural.

Classical Field Theory	ℓ^1 Cohomological Framework
Action $\mathcal{S}$	$\|\omega - d_0 s\|_1$
Field	Section s
Curvature	Holonomy $\mathrm{Hol}(C)$
Gauge invariance	$s \sim s + f$ (quotient structure)
Field equation	$d_0^*\phi = 0$
Conserved current	Divergence-free ϕ

Continuum Version. Under scaling, the conservation law becomes $\nabla \cdot \phi = 0$ with ϕ the continuum dual flux. This is the classical conservation equation structure.

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5. The Distinguishability Bound (The Bridge to Physics)

5.1 The Distinguishability Bound Theorem

The results of Papers 000–004 establish conditional ℓ^1 rigidity for additive defect diagnostics under the stated observer assumptions, and show how the framework extends to Schatten-1 trace geometry once the operator-valued local gauge is selected. This permits a structural comparison with standard distance metrics in information theory; it is not an unconditional derivation of quantum distinguishability.

> **Theorem 5.1 (Distinguishability Bound). (Layer B)** Let $(\rho_x)_{x\in P}$ be an operator-valued presheaf section over a path graph connecting nodes a and b, with uniform local Hilbert spaces $\mathcal{H}_x = \mathcal{H}$ for all x, and let $I(s) = \sum_e \|\delta_e(s)\|_1$ be the ℓ^1 coboundary defect. Let $R_{b,a} = r_{1,0} \circ r_{2,1} \circ \cdots \circ r_{m,m-1}$ denote the composed restriction (parallel transport) from b to a. Then the quantum trace distance

satisfies:

$$T(\rho_a, R_{b,a}(\rho_b)) \leq \frac{1}{2} I(s)$$

Proof Sketch. By telescoping and the contractivity of each restriction map: $\|R_{b,a}(\rho_b) - \rho_a\|_1 \leq \sum_i \|r_{i+1,i}(\rho_{i+1}) - \rho_i\|_1 = \sum_{e \in \mathrm{path}(a,b)} \|\delta_e\|_1 = I(s)$. Noting that trace distance is precisely half the trace norm yields the result. When all Hilbert spaces are identical and restriction maps are the identity, this reduces to the standard triangle inequality $T(\rho_a, \rho_b) \leq \frac{1}{2} I(s)$. □

This theorem provides a structural link between the framework and operational distinguishability in quantum information, contingent on identifying presheaf sections with physical quantum states. Under the operator-valued presheaf interpretation and contractive restriction maps, the path defect sum bounds the trace distance between endpoint states after transport along that path. Furthermore, returning to the Hodge cohomology of Paper 003, the quotient norm $\Phi([\delta])$ bounds the *minimal residual trace-norm defect within the declared quotient model* (irreducible correlation) that remains after all local gauge transformations are exhausted.

5.2 Connection to the Fermion Sign Problem (Layer C — Conjectural)

The Schatten-1 extension (Theorem 3.1) suggests a structural connection to the fermion sign problem. When the presheaf stalks carry fermionic statistics, the restriction maps $r_{b,a}$ involve antisymmetrization, and the coboundary defect δ_e would measure the failure of local fermionic states to glue consistently. Whether the trace norm of this defect can be rigorously identified with sign frustration requires formal development beyond this paper's scope [11].

If this connection holds, it would provide a structural interpretation: the fermion sign problem would be a manifestation of ℓ^1 gauge frustration on operator-valued presheaves, rather than an artifact of computational representation. However, this remains conjectural and requires explicit construction of the fermionic presheaf and verification that the coboundary defect captures the relevant sign structure.

5.3 Connection to Coupling Constraints and POVM Equivalence (Layer C — Quarantined Conjectural Boundary)

Subsequent physics work frequently maps structural edge diagnostics to POVM limit measurements operating over continuous phase-symmetric state spaces (L^2_{RMS}). This mapping is subject to Conditional Minimality. The L^2 Hilbert phase envelope is selected only when supplementary assumptions (complex amplitudes, bilinear cross terms, reversible unitary evolutions) are explicitly injected over tracking limits. The formal equivalence between presheaf topological mapping gaps and explicit Positive Operator-Valued Measure behavior remains open. The structural engine does *not* prove quantum mechanics universally; it

gives a trace/Schatten-1 obstruction geometry only after the operator-valued observer has selected the trace-distance gauge.

Whether the structural emergence of wave mechanics, the Born rule, and subsequent gauge coupling constants formally descends entirely autonomously from these trace constraints remains deeply speculative.

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6. Discussion

6.1 What This Paper Proves

This paper extends the foundational quad (Papers 000–003) in five directions:

1. **Functorial aggregation is contractive** (Theorem 2.1): under explicit contractive pushforward assumptions, coarse-graining cannot increase the ℓ^1 defect norm. Defects may still be hidden by collapse or cancellation.
2. **The operator-valued extension is conditional** (Theorem 3.1): once the trace/Schatten-1 local gauge is selected, the edge-additive extension is $\ell^1(E; \|\cdot\|_1)$. CPTP monotonicity alone is insufficient.
3. **The Local ℓ^1 Primal–Dual Iterator characterizes defect fixed points** (Definition 4.1, Theorem 4.2): a local ℓ^1 primal-dual iterator whose KKT-feasible fixed points are ℓ^1-minimal representatives. Edge-wise descent remains heuristic (Conjecture 4.4), and monotonic global descent is not proved.
4. **Certified cycle-supported fixed points and interaction heuristics** (Proposition 4.5, Heuristic 4.6, Proposition 4.7): cycle-supported dual-certified residuals are fixed points under stated invariance assumptions; cycle interaction and propagation behavior are formal/local-update heuristics, not convergence theorems.
5. **A quotient variational structure exists for fixed observed fields:** the finite ℓ^1 quotient action and KKT dual condition are proved; the continuum, curvature, holonomy, and Noether-style interpretations are formal schemas.

6.2 What This Paper Does Not Prove

The conjectural elements (Layer C) are explicitly marked:

- The continuum limit (Theorem 4.8) is stated as a formal scaling limit; full Γ-convergence is not proved.
- The geometry emergence result (Theorem 4.11) invokes established results from geometric measure theory under standard BV regularity assumptions but does not re-derive them.
- The connections to the Standard Model gauge groups remain structural observations pending explicit derivation.

- The derivation chain to quantum mechanics (Section 5.1) contains both established and open steps.
- Monotonic descent of the iterator (Conjecture 4.3) is not proved; the convergence analysis for the three-step primal–dual variant is deferred to subsequent work.
- The finite-speed propagation (Corollary 4.2) establishes causal ordering but does not imply Lorentz invariance, an invariant propagation speed, or relativistic frame equivalence; whether additional symmetry constraints on the ℓ^1 action can force an invariant speed remains open.

The conditionality of all claims is preserved. The ℓ^1 framework is selected within the axiom class of Papers 001–002. The extensions proved here hold given that ℓ^1 framework and their additional local-gauge assumptions. The conjectural elements indicate where the program requires further formal development.

6.3 Position in the Program

Papers 000–003 answer: *What defect metric is selected under the stated additive observer axioms?* (weighted ℓ^1, with uniform ℓ^1 only under edge-uniformity.)

This paper answers: *What is its static and variational structure?* (contractive under aggregation, conditionally extended to operators, governed by a quotient action whose symmetries generate conserved flux, with emergent connection-curvature-holonomy structure and a formal continuum limit recovering total variation flow.)

Paper 005 inherits: *How does local dynamics realize this action?* (The Autopoietic Iterator as the local engine driving causal poset construction, propagation, and structural emergence.)

The program is falsifiable at every joint: if any proved theorem is incorrect, or if any conjectured step fails, the downstream structure requires revision at that point. The separation of established results (Layers A, B) from conjectures (Layer C) is maintained to make this falsification structure transparent.

6.4 The Scope of Geometric Emergence

The ℓ^1 framework does not prescribe the ambient geometric substrate—the topology, dimensionality, or embedding of the underlying graph or poset are inputs, not outputs. Rather, the framework determines the variational structure governing defect minimization, which in turn forces all nontrivial structure to concentrate on minimal geometric carriers. In the continuum limit, this yields codimension-1 interfaces evolving under curvature-driven dynamics. Geometry in this sense is not imposed a priori but emerges as the support of irreducible defect under the ℓ^1 action.

Concretely, the framework derives:

- **Where structure lives:** on minimal-measure carriers (cycles $\to$ interfaces)
- **How structure behaves:** curvature-driven evolution, finite-speed propagation
- **What counts as irreducible:** gauge-invariant cohomological content ($\Phi([\delta])$)

It does not derive:

- The ambient dimension or topology of the substrate
- A Riemannian metric on the background space
- The specific graph connectivity or embedding

This distinction positions the framework as a theory of **geometric content** (what structures exist and how they interact) rather than **geometric background** (what space they inhabit). The substrate is given; the induced variational structure on that substrate is analyzed. The framework derives causal structure first (finite-speed propagation, causal ordering from variational descent); metric structure—including Lorentz symmetry, invariant speed, and observer equivalence—remains an open emergence problem requiring additional symmetry constraints on the action or the graph ensemble.

6.5 Open Problems: Emergent Metric and Homogenization

The cone structure identified in Remark 4.3 and the defect-induced path cost $d_\delta(x, y) = \inf_\gamma \sum_{e\in\gamma} |\delta(e)|$ define a candidate emergent metric. Three open questions determine whether this candidate can bridge to physical geometry:

1. **Homogenization under aggregation.** While the defect-induced metric is generically anisotropic at microscopic scales, the combination of ℓ^1 minimization (which penalizes redundant directional concentration) and functorial aggregation (Theorem 2.1, which preserves total defect under coarse-graining) suggests a homogenization mechanism. *Open problem:* Does repeated coarse-graining of generic ℓ^1 defect configurations converge to an approximately isotropic metric in distribution, for sufficiently symmetric or ergodic graph ensembles?

2. **Invariant speed.** The dual feasibility constraint $\|\phi\|_\infty \leq 1$ imposes a uniform flux bound, yielding a maximal propagation rate. *Open problem:* Under what symmetry conditions on the ℓ^1 action does this maximal rate become invariant under admissible transformations?

3. **Metric–dynamics coupling.** The defect-induced metric d_δ depends on the field configuration, so geometry and dynamics are coupled. *Open problem:* Does this coupling admit a consistent continuum limit in which the emergent metric satisfies a geometric evolution equation (analogous to Ricci flow or Einstein equations)?

A positive answer to (1) would establish a serious bridge between the ℓ^1 framework

and spacetime structure. A counterexample—a stable, strongly anisotropic defect configuration resistant to aggregation—would bound the geometric ambitions of the program. This constitutes the critical falsification test for the physics-level extension of the framework.

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Changelog

v5.0 (May 2026)

- Added cross-reference to Paper 000 v004 [13] (observer passivity, contextual congruence, reflective completion, lift contracts).
- The operator-valued extension (Theorem 3.1), fixed-point characterization (Theorem 4.2), and gauge-invariant action (Theorem 4.14) all operate within the observer-relative diagnostic framework formalized in [13]. Observer passivity (O1) constrains downstream processing of the quotient obstruction Φ computed by the ℓ^1 primal-dual structure.

v4.0 (April 2026)

- Audit-hardened pre-print. Layer A/B/C classification finalized.